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Enhanced color gamut performance of a liquid crystal display

Abstract

An information handling system determines an expansion ratio based on a highest grayscale value of a raw image and a maximum grayscale value, and modifies at least one grayscale value associated with the raw image prior to display at a display panel, wherein the modifying of the grayscale value is based on the expansion ratio, and wherein the highest grayscale value is updated with the maximum grayscale value. The system adjusts backlight intensity based on a gamma correction curve to adjust a pixel transmittance of the display panel.

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Background/Summary

FIELD OF THE DISCLOSURE

(1) The present disclosure generally relates to information handling systems, and more particularly relates to enhanced color gamut performance of a liquid crystal display.

BACKGROUND

(2) As the value and use of information continues to increase, individuals and businesses seek additional ways to process and store information. One option is an information handling system. An information handling system generally processes, compiles, stores, or communicates information or data for business, personal, or other purposes. Technology and information handling needs and requirements can vary between different applications. Thus, information handling systems can also vary regarding what information is handled, how the information is handled, how much information is processed, stored, or communicated, and how quickly and efficiently the information can be processed, stored, or communicated. The variations in information handling systems allow information handling systems to be general or configured for a specific user or specific use such as financial transaction processing, airline reservations, enterprise data storage, or global communications. In addition, information handling systems can include a variety of hardware and software resources that can be configured to process, store, and communicate information and can include one or more computer systems, graphics interface systems, data storage systems, networking systems, and mobile communication systems. Information handling systems can also implement various virtualized architectures. Data and voice communications among information handling systems may be via networks that are wired, wireless, or some combination.

SUMMARY

(3) An information handling system determines an expansion ratio based on a highest grayscale value of a raw image and a maximum grayscale value, and modifies at least one grayscale value associated with the raw image prior to display at a display panel, wherein the modifying of the grayscale value is based on the expansion ratio, and wherein the highest grayscale value is updated with the maximum grayscale value. The system adjusts backlight intensity based on a gamma correction curve to adjust a pixel transmittance of the display panel.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) It will be appreciated that for simplicity and clarity of illustration, elements illustrated in the Figures are not necessarily drawn to scale. For example, the dimensions of some elements may be exaggerated relative to other elements. Embodiments incorporating teachings of the present disclosure are shown and described with respect to the drawings herein, in which:

(2) FIG. **1** is a block diagram illustrating an information handling system according to an embodiment of the present disclosure;

(3) FIG. **2** is a block diagram of a display that is configured for enhanced color gamut performance, according to an embodiment of the present disclosure;

(4) FIG. **3** is a block diagram of a display with a display panel that has been divided into local area units, according to an embodiment of the present disclosure;

(5) FIG. **4** is a block diagram of modified display settings for enhanced color gamut performance of a liquid crystal display, according to an embodiment of the present disclosure;

(6) FIGS. **5** and **6** are diagrams of plots of gamma curves for a display, according to an embodiment of the present disclosure;

(7) FIG. **7** is a diagram of grayscale value and backlight intensity conversion for an enhanced color gamut performance of liquid crystal display, according to an embodiment of the present disclosure; and

(8) FIG. **8** is a flowchart of a method for enhanced color gamut performance of a liquid crystal display; according to an embodiment of the present disclosure.

(9) The use of the same reference symbols in different drawings indicates similar or identical items.

DETAILED DESCRIPTION OF THE DRAWINGS

(10) The following description in combination with the Figures is provided to assist in understanding the teachings disclosed herein. The description is focused on specific implementations and embodiments of the teachings and is provided to assist in describing the teachings. This focus should not be interpreted as a limitation on the scope or applicability of the teachings.
(11) FIG. **1** illustrates an embodiment of an information handling system **100** including processors **102** and **104**, a chipset **110**, a memory **120**, a graphics adapter **130** connected to a video display **134**, a non-volatile RAM (NV-RAM) **140** that includes a basic input and output system/extensible firmware interface (BIOS/EFI) module **142**, a disk controller **150**, a hard disk drive (HDD) **154**, an optical disk drive **156**, a disk emulator **160** connected to a solid-state drive (SSD) **164**, an input/output (I/O) interface **170** connected to an add-on resource **174** and a trusted platform module (TPM) **176**, a network interface **180**, and a baseboard management controller (BMC) **190**. Processor **102** is connected to chipset **110** via processor interface **106**, and processor **104** is connected to the chipset via processor interface **108**. In a particular embodiment, processors **102** and **104** are connected via a high-capacity coherent fabric, such as a HyperTransport link, a QuickPath Interconnect, or the like. Chipset **110** represents an integrated circuit or group of integrated circuits that manage the data flow between processors **102** and **104** and the other elements of information handling system **100**. In a particular embodiment, chipset **110** represents a pair of integrated circuits, such as a northbridge component and a southbridge component. In another embodiment, some or all the functions and features of chipset **110** are integrated with one or more of processors **102** and **104**.
(12) Memory **120** is connected to chipset **110** via a memory interface **122**. An example of memory interface **122** includes a Double Data Rate (DDR) memory channel and memory **120** represents one or more DDR Dual In-Line Memory Modules (DIMMs). In a particular embodiment, memory interface **122** represents two or more DDR channels. In another embodiment, one or more of processors **102** and **104** include a memory interface that provides a dedicated memory for the processors. A DDR channel and the connected DDR DIMMs can be in accordance with a particular DDR standard, such as a DDR3 standard, a DDR4 standard, a DDR5 standard, or the like.
(13) Memory **120** may further represent various combinations of memory types, such as Dynamic Random Access Memory (DRAM) DIMMs, Static Random Access Memory (SRAM) DIMMs, non-volatile DIMMs (NV-DIMMs), storage class memory devices, Read-Only Memory (ROM) devices, or the like. Graphics adapter **130** is connected to chipset **110** via a graphics interface **132** and provides a video display output **136** to a video display **134**. An example of a graphics interface **132** includes a Peripheral Component Interconnect-Express (PCIe) interface and graphics adapter **130** can include a four-lane (×4) PCIe adapter, an eight-lane (×8) PCIe adapter, a 16-lane (×16) PCIe adapter, or another configuration, as needed or desired. In a particular embodiment, graphics adapter **130** is provided down on a system printed circuit board (PCB). Video display output **136** can include a Digital Video Interface (DVI), a High-Definition Multimedia Interface (HDMI), a DisplayPort interface, or the like, and video display **134** can include a monitor, a smart television, an embedded display such as a laptop computer display, or the like.
(14) NV-RAM **140**, disk controller **150**, and I/O interface **170** are connected to chipset **110** via an I/O channel **112**. An example of I/O channel **112** includes one or more point-to-point PCIe links between chipset **110** and each of NV-RAM **140**, disk controller **150**, and I/O interface **170**. Chipset **110** can also include one or more other I/O interfaces, including a PCIe interface, an Industry Standard Architecture (ISA) interface, a Small Computer Serial Interface (SCSI) interface, an Inter-Integrated Circuit (I2C) interface, a System Packet Interface (SPI), a Universal Serial Bus (USB), another interface, or a combination thereof. NV-RAM **140** includes BIOS/EFI module **142** that stores machine-executable code (BIOS/EFI code) that operates to detect the resources of information handling system **100**, to provide drivers for the resources, to initialize the resources, and to provide common access mechanisms for the resources. The functions and features of

BIOS/EFI module **142** will be further described below.

(15) Disk controller **150** includes a disk interface **152** that connects the disc controller to a hard disk drive (HDD) **154**, to an optical disk drive (ODD) **156**, and to disk emulator **160**. An example of disk interface **152** includes an Integrated Drive Electronics (IDE) interface, an Advanced Technology Attachment (ATA) such as a parallel ATA (PATA) interface or a serial ATA (SATA) interface, a SCSI interface, a USB interface, a proprietary interface, or a combination thereof. Disk emulator **160** permits SSD **164** to be connected to information handling system **100** via an external interface **162**. An example of external interface **162** includes a USB interface, an institute of electrical and electronics engineers (IEEE) 1394 (Firewire) interface, a proprietary interface, or a combination thereof. Alternatively, SSD **164** can be disposed within information handling system **100**.

(16) I/O interface **170** includes a peripheral interface **172** that connects the I/O interface to add-on resource **174**, to TPM **176**, and to network interface **180**. Peripheral interface **172** can be the same type of interface as I/O channel **112** or can be a different type of interface. As such, I/O interface **170** extends the capacity of I/O channel **112** when peripheral interface **172** and the I/O channel are of the same type, and the I/O interface translates information from a format suitable to the I/O channel to a format suitable to the peripheral interface **172** when they are of a different type. Add-on resource **174** can include a data storage system, an additional graphics interface, a network interface card (NIC), a sound/video processing card, another add-on resource, or a combination thereof. Add-on resource **174** can be on a main circuit board, on separate circuit board, or add-in card disposed within information handling system **100**, a device that is external to the information handling system, or a combination thereof.

(17) Network interface **180** represents a network communication device disposed within information handling system **100**, on a main circuit board of the information handling system, integrated onto another component such as chipset **110**, in another suitable location, or a combination thereof. Network interface **180** includes a network channel **182** that provides an interface to devices that are external to information handling system **100**. In a particular embodiment, network channel **182** is of a different type than peripheral interface **172** and network interface **180** translates information from a format suitable to the peripheral channel to a format suitable to external devices.

(18) In a particular embodiment, network interface **180** includes a NIC or host bus adapter (HBA), and an example of network channel **182** includes an InfiniBand channel, a Fibre Channel, a Gigabit Ethernet channel, a proprietary channel architecture, or a combination thereof. In another embodiment, network interface **180** includes a wireless communication interface, and network channel **182** includes a Wi-Fi channel, a near-field communication (NFC) channel, a Bluetooth® or Bluetooth-Low-Energy (BLE) channel, a cellular based interface such as a Global System for Mobile (GSM) interface, a Code-Division Multiple Access (CDMA) interface, a Universal Mobile Telecommunications System (UMTS) interface, a Long-Term Evolution (LTE) interface, or another cellular based interface, or a combination thereof. Network channel **182** can be connected to an external network resource (not illustrated). The network resource can include another information handling system, a data storage system, another network, a grid management system, another suitable resource, or a combination thereof.

(19) BMC **190** is connected to multiple elements of information handling system **100** via one or more management interface **192** to provide out of band monitoring, maintenance, and control of the elements of the information handling system. As such, BMC **190** represents a processing device different from processor **102** and processor **104**, which provides various management functions for information handling system **100**. For example, BMC **190** may be responsible for power management, cooling management, and the like. The term BMC is often used in the context of server systems, while in a consumer-level device, a BMC may be referred to as an embedded controller (EC). A BMC included in a data storage system can be referred to as a storage enclosure

processor. A BMC included at a chassis of a blade server can be referred to as a chassis management controller and embedded controllers included at the blades of the blade server can be referred to as blade management controllers. Capabilities and functions provided by BMC **190** can vary considerably based on the type of information handling system. BMC **190** can operate in accordance with an Intelligent Platform Management Interface (IPMI). Examples of BMC **190** include an Integrated Dell® Remote Access Controller (iDRAC).

(20) Management interface **192** represents one or more out-of-band communication interfaces between BMC **190** and the elements of information handling system **100**, and can include an Inter-Integrated Circuit (I2C) bus, a System Management Bus (SMBUS), a Power Management Bus (PMBUS), a Low Pin Count (LPC) interface, a serial bus such as a Universal Serial Bus (USB) or a Serial Peripheral Interface (SPI), a network interface such as an Ethernet interface, a high-speed serial data link such as a PCIe interface, a Network Controller Sideband Interface (NC-SI), or the like. As used herein, out-of-band access refers to operations performed apart from a BIOS/operating system execution environment on information handling system **100**, that is apart from the execution of code by processors **102** and **104** and procedures that are implemented on the information handling system in response to the executed code.

(21) BMC **190** operates to monitor and maintain system firmware, such as code stored in BIOS/EFI module **142**, option ROMs for graphics adapter **130**, disk controller **150**, add-on resource **174**, network interface **180**, or other elements of information handling system **100**, as needed or desired. In particular, BMC **190** includes a network interface **194** that can be connected to a remote management system to receive firmware updates, as needed or desired. Here, BMC **190** receives the firmware updates, stores the updates to a data storage device associated with the BMC, transfers the firmware updates to NV-RAM of the device or system that is the subject of the firmware update, thereby replacing the currently operating firmware associated with the device or system, and reboots information handling system, whereupon the device or system utilizes the updated firmware image.

(22) BMC **190** utilizes various protocols and application programming interfaces (APIs) to direct and control the processes for monitoring and maintaining the system firmware. An example of a protocol or API for monitoring and maintaining the system firmware includes a graphical user interface (GUI) associated with BMC **190**, an interface defined by the Distributed Management Taskforce (DMTF) (such as a Web Services Management (WSMan) interface, a Management Component Transport Protocol (MCTP) or, a Redfish® interface), various vendor defined interfaces (such as a Dell EMC Remote Access Controller Administrator (RACADM) utility, a Dell EMC OpenManage Enterprise, a Dell EMC OpenManage Server Administrator (OMSS) utility, a Dell EMC OpenManage Storage Services (OMSS) utility, or a Dell EMC OpenManage Deployment Toolkit (DTK) suite), a BIOS setup utility such as invoked by a “F2” boot option, or another protocol or API, as needed or desired.

(23) In a particular embodiment, BMC **190** is included on a main circuit board (such as a baseboard, a motherboard, or any combination thereof) of information handling system **100** or is integrated onto another element of the information handling system such as chipset **110**, or another suitable element, as needed or desired. As such, BMC **190** can be part of an integrated circuit or a chipset within information handling system **100**. An example of BMC **190** includes an iDRAC, or the like. BMC **190** may operate on a separate power plane from other resources in information handling system **100**. Thus BMC **190** can communicate with the management system via network interface **194** while the resources of information handling system **100** are powered off. Here, information can be sent from the management system to BMC **190** and the information can be stored in a RAM or NV-RAM associated with the BMC. Information stored in the RAM may be lost after power-down of the power plane for BMC **190**, while information stored in the NV-RAM may be saved through a power-down/power-up cycle of the power plane for the BMC.

(24) Information handling system **100** can include additional components and additional busses,

not shown for clarity. For example, information handling system **100** can include multiple processor cores, audio devices, and the like. While a particular arrangement of bus technologies and interconnections is illustrated for the purpose of example, one of skill will appreciate that the techniques disclosed herein are applicable to other system architectures. Information handling system **100** can include multiple central processing units (CPUs) and redundant bus controllers. One or more components can be integrated together. Information handling system **100** can include additional buses and bus protocols, for example, I2C and the like. Additional components of information handling system **100** can include one or more storage devices that can store machine-executable code, one or more communications ports for communicating with external devices, and various input and output (I/O) devices, such as a keyboard, a mouse, and a video display.

(25) For purposes of this disclosure, information handling system **100** can include any instrumentality or aggregate of instrumentalities operable to compute, classify, process, transmit, receive, retrieve, originate, switch, store, display, manifest, detect, record, reproduce, handle, or utilize any form of information, intelligence, or data for business, scientific, control, entertainment, or other purposes. For example, information handling system **100** can be a personal computer, a laptop computer, a smartphone, a tablet device or other consumer electronic device, a network server, a network storage device, a switch, a router, or another network communication device, or any other suitable device and may vary in size, shape, performance, functionality, and price. Further, information handling system **100** can include processing resources for executing machine-executable code, such as processor **102**, a programmable logic array (PLA), an embedded device such as a System-on-a-Chip (SoC), or other control logic hardware. Information handling system **100** can also include one or more computer-readable media for storing machine-executable code, such as software or data.

(26) One issue with liquid crystal displays is that their color gamut typically narrows as the grayscale becomes darker. Liquid crystal displays use a backlight as a source for light emission and liquid crystals control the light's transmittance. Due to structural limitations, these displays generally cannot completely block the backlight resulting in light leakage. The light leakage may result in result in color purity degradation such as when red, green, and blue colors are displayed individually. For example when the red color is displayed, the green and blue colors should not emit light. However, the light leakage typically occurs in the green and blue colors and the light emitted by the green and blue colors can mix. This causes color purity and coordinates distortion of the red color.

(27) Generally, the color gamut degradation begins at around sixty-four grayscale and becomes almost zero at around four grayscale value. This means that the color may get distorted when the liquid crystal display device displays red, green, and blue colors below the sixty-four grayscale value. In this example the lower the grayscale, the red color coordinates get closer to the white color coordinates which causes a reduction in color space. This in turn may cause difficulty in distinguishing between the red, green, and blue colors. To address these and other concerns, the present disclosure provides a system and method for enhanced color gamut performance in a liquid crystal display.

(28) FIG. **2** shows a display **200** that is configured for enhanced color gamut performance. Display **200** includes a scaler **210**, a display panel **220**, a power board **230**, a backlight unit **240**, and a display port **250**. Scaler **210** includes a color gamut enhancement module **215**. Scaler **210**, display panel **220**, power board **230**, backlight unit **240**, and display port **250** may be communicatively coupled with each other. The components of display **200** may be implemented in hardware, software, firmware, or any combination thereof. In addition, the components shown are not drawn to scale, and display **200** may include additional or fewer components. Further, connections between components may be omitted for descriptive clarity.

(29) Display **200**, which is similar to video display **134** of FIG. **1**, maybe a liquid crystal display that is used to display information to a user. Display **200** may be integrated into an information

handling system. In another embodiment, display **200** may be an external display device, such as a display monitor. Scaler **210** may be configured to provide support for inputs from different image formats, such as video graphics array, display port, digital visual interface, high-definition multimedia interface, etc. In one example, scaler **210** may receive raw image data from display port **250**. Scaler **210** may process the raw image data for resolution, refresh rate, and color optimization. Also, scaler **210** may decode the received image format into a digital pixel format. In addition, scaler **210** may be configured to convert the resolution of the received image to a resolution of display **200**. For example, scaler **210** may upscale or downscale the resolution of the received image to accommodate the resolution of display **200**. As the received image is upscaled or downscaled, scaler **210** may process the received image to adjust for refresh rate and color.

(30) Color gamut enhancement module **215** may include an algorithm to enhance the color gamut of display **200** by minimizing light leakage by minimizing backlight intensity and maximizing liquid crystal transmittance. The algorithm may perform a conversion of the current and/or voltage provided to backlight unit **240** and adjust grayscale values of the image data based on a gamma correction curve which is discussed in further detail in FIG. **4**. In particular, color gamut enhancement module **215** may convert the greyscale values of a received image prior to outputting a final image for display panel **220**.

(31) Color gamut enhancement module **215** may also be configured to update a pulse width modulation (PWM) signal value it provides to power board **230**. The PWM signal value may be used by power board **230** to determine a current and/or voltage provided to backlight unit **240**. The current may be used to adjust or drive the backlight intensity provided by backlight unit **240**, such as to dim or increase the brightness of backlight unit **240**. For example, the higher the current output to backlight unit **240**, the brighter the light output of backlight unit **240**. Conversely, the lower the current output to backlight unit **240**, the dimmer the light output of backlight unit **240**.

(32) In another embodiment, the brightness provided by backlight unit **240** may be controlled by adjusting the on/off ratio. For example, to achieve 70% brightness, the digital signal may be kept on for 70% of the time and off for 30% of the time the display panel is on. This is typically done rapidly, such as a certain frequency per second. In one embodiment, color gamut enhancement module **215** may be configured to perform the conversions in real-time. In another embodiment, a set of look-up tables may be generated during the production of display **200** and stored in a non-volatile memory that is accessible by color gamut enhancement module **215**. The set of lookup tables may include grayscale values and backlight intensity values that color gamut enhancement module **215** may use during the conversion.

(33) Power board **230** may be configured to provide power to backlight unit **240**. Power board **230** may receive the PWM signal from scaler **210** and can output current and/or voltage to backlight unit **240** based on the PWM signal. Backlight unit **240** may be configured to illuminate the liquid crystal material to provide a contrast between the light transmissive and opaque portions of display **200**. Backlight unit **240** may be a cool cathode fluorescent light, several light-emitting diodes, an electroluminescent panel, or any other suitable device. Backlight unit **240** may be an edge-type backlight unit or a direct-type backlight unit.

(34) Display panel **220** may be a liquid crystal display that can modulate light to create images using selectively transmissive and opaque portions of the display by passing an electrical current through a liquid crystal material. Display **114** may have a display surface that includes multiple pixels. Each pixel in display **200** may include liquid crystal molecules suspended between two transparent electrodes that are in turn sandwiched between two polarizing filters whose axes of transmission may be perpendicular to each other. By applying voltage to the transparent electrodes over each pixel, the corresponding liquid crystal molecules may allow varying degrees of light to pass through the polarizing filters. Each pixel may be composed of individual red, green, blue (RGB), and/or other color sub-pixels. In addition, each pixel is assigned a value of grayscale level between 0 and 255.

(35) Those of ordinary skill in the art will appreciate that the configuration, hardware, and/or software components of display **200** depicted in FIG. **2** may vary. For example, the illustrative components within display **200** are not intended to be exhaustive but rather are representative to highlight components that can be utilized to implement aspects of the present disclosure. For example, other devices and/or components may be used in addition to or in place of the devices/components depicted. The depicted example does not convey or imply any architectural or other limitations with respect to the presently described embodiments and/or the general disclosure. In the discussion of the figures, reference may also be made to components illustrated in other figures for continuity of the description.
(36) FIG. **3** shows display **200** with a display panel that has been divided into a plurality of local area units, wherein each local area unit may be subdivided further into sub-units. Each subunit may include one or more pixels. Each local area unit and sub-unit may be of equal size. In this example, display panel **220** is divided into 35 local area units while each local area unit, such as a local area unit **305** is divided into nine subunits, **315-1**, **315-2**, through **315-9**. Each local area unit may be associated with grayscale settings which are based on the raw image data. For example, local area unit **305** includes grayscale settings **320** that further include various grayscale values, wherein each subunit may be associated with a particular grayscale value. The grayscale value may be the same or different from the grayscale values of other subunits. In this example, subunit **315-1** has a 15 grayscale, **315-2** has 11 grayscale, and subunit **315-9** has 0 grayscale, and so on. In addition, backlight unit **240** disposed behind display panel **220** may also be divided into local area units and subunits similarly to display panel **220**, wherein each local area unit of display panel **220** has a corresponding backlight local area unit.
(37) FIG. **4** shows modified display settings for enhanced color gamut performance of a liquid crystal display. The modified display settings include display settings **440** and pixel transmittance values **450**. Display settings **440** include backlight intensity settings **425** for a local area unit of backlight unit **240** of FIG. **2**. Backlight intensity settings **425** may be modified from a static backlight intensity setting of 100%. A local area unit of a backlight unit may be disposed on the back side of local area unit **305** of display panel **220**. Display settings **400** also include grayscale settings **420** for local area unit **305** which can be modified settings based on grayscale settings **320** of FIG. **3**. The conversion algorithm for converting the backlight intensity settings and the grayscale settings from their initial values is discussed below.
(38) Converting grayscale settings **320** to grayscale settings **420** includes replacing the highest grayscale value of grayscale settings **320** to a maximum grayscale value of 255G. The conversion is applied to each subunit of each local area unit. This may be done to maximize liquid crystal transmittance as depicted in pixel transmittance values **450** and to calculate an expansion ratio. The expansion ratio may relate to an expansion liquid crystal operation range. For example, typically a gramma curve, such as a 2.2 gamma curve is applied to raw image data based on the grayscale value of each subunit. In this example, the grayscale value of subunit **315-1** is 15G, which is also the highest grayscale value in local area unit **305**. This grayscale value may be increased to the maximum grayscale value of 255G.
(39) Accordingly, gamma correction can then be applied from a grayscale range of 15G to 255G as depicted in FIG. **5** and FIG. **6**. Thus, the expansion ratio is 255G/15G which equals 17. The expansion ratio can be used in converting each remaining grayscale value of each subunit of the local area units of display **200**. For example, the grayscale value of subunit **315-2** may be converted from 11G to 187G, wherein 187G is calculated from 11G*17. Similarly, the grayscale value of subunit **315-0** may be converted from 0G to 0G, wherein 0G is calculated from 0G*17.
(40) The current for backlight light elements associated with each subunit of the local area units of backlight unit **240** may be adjusted or converted. The conversion may include dimming the backlight intensity for that subunit from a typical static backlight intensity of 100%. The conversion may be based on a formula: (grayscale value of the subunit/maximum grayscale value)

{circumflex over ( )}2.2, wherein 2.2 is based on a conventional gamma curve, although other gamma curves may be used. Accordingly, the subunit with the highest grayscale value may also have the highest backlight intensity and highest pixel transmittance as depicted in backlight intensity settings **425** and pixel transmittance values **450**. For example, for subunit **315-1**, the current for backlight elements for a particular subunit may be adjusted such that its backlight intensity is equal to 0.196%, which is calculated from (15/255){circumflex over ( )}2.2. Accordingly, the backlight intensity of the rest of the subunits may be adjusted such that the backlight intensity of each subunit is equal to 0.196%, which is based on the subunit with the highest grayscale.

(41) The combination of the backlight intensity settings **425** and grayscale settings **430** may result in modified grayscale settings **420** which are similar to initial grayscale settings **320**. Further, the combination of the backlight intensity settings **425** and grayscale settings **420** may result in pixel transmittance values **450**. Pixel transmittance values **450** may also be calculated based on the gamma curve. Assuming that 225G results in 100% pixel transmittance and 0G results in 0% pixel transmittance, then calculating the pixel transmittance of the subunits may be based on its (grayscale setting/maximum grayscale setting){circumflex over ( )}2.2 gamma curve. For example, pixel transmittance **435-1** may be determined to be 100%, and pixel transmittance **435-9** to be 0%. In addition, pixel transmittance **435-2** may be determined to be (185G/255G){circumflex over ( )}2.2 which results in 50.54%. The modified backlight intensity, grayscale settings, and pixel transmittance should result in an improvement of the color gamut. In addition, because of adjustments or reductions to the current provided to backlight unit **240**, the power consumption of display **200** may be reduced.

(42) FIG. **5** shows a plot **500** of a gamma curve **510** for display **200** of FIG. **3**. Gamma curve **510** indicates pixel transmittance in percentages as a function of pixel grayscale level which can vary from 0 to 255. In this example, gamma curve **610** is a 2.2 gamma curve applied to a grayscale range of 0 to approximately 15G, which is the highest grayscale for local area unit **305** of FIG. **3**.

(43) FIG. **6** shows a plot **600** of a gamma curve **610** for display **200** of FIG. **3**, wherein gamma curve **610** is a 2.2 gamma curve which is applied to an expanded grayscale range of 0 to approximately 255G, which is the maximum grayscale value. Accordingly, the expansion ratio may be based on the highest grayscale/maximum grayscale. In this example, the expansion ratio of 15G/255 is equal to 17.

(44) FIG. **7** shows a grayscale value and backlight intensity conversion for an enhanced color gamut performance of liquid crystal display. This example includes display settings **715** and modified display settings **740**. Display settings **715** may be based on gamma curve **510** of FIG. **5** while display settings **740** can be based on gamma curve **610** of FIG. **6**. Display settings **715** include backlight intensity settings **705**, grayscale settings **320**, and grayscale settings **710**. Backlight intensity settings **705** which are set to 100% for a local area unit of backlight unit **240** of FIG. **2**. In this example, the backlight intensity of 100% is static and of the same value regardless of grayscale settings. The associated local area unit may be disposed on the back side of local area unit **305** of display panel **220**. Grayscale settings **320** of local area unit **305** include various grayscale settings for each subunit. The combination results in grayscale settings **710**, which can be desired grayscale settings.

(45) Color gamut enhancement module **215** may convert display settings **715** to display settings **740** which are similar to display settings **440** of FIG. **4**. Display settings **740** include backlight intensity settings **725**, grayscale settings **720**, and grayscale settings **730**. Backlight intensity settings **725** are similar to backlight intensity settings **425**. Grayscale settings **720** are similar to grayscale settings **420** and grayscale settings **730** are similar to grayscale settings **420**. Backlight intensity settings **725** include different backlight intensities for each subunit of a local area unit. This is instead of having 100% backlight intensity for each subunit. The conversion from backlight intensity settings **705** to backlight intensity settings **725** may be calculated as shown in FIG. **4**.

Similarly, the conversion from grayscale settings **320** to grayscale settings **720** was shown in FIG. **4**. Accordingly, the combination of backlight intensity settings **725** and grayscale settings **720** results in grayscale settings **730** which is similar to grayscale settings **710**.
(46) FIG. **8** shows a flowchart of a method **800** for enhanced color gamut performance of a liquid crystal display. Method **800** may be performed by any suitable component including, but not limited to, scaler **210**, display panel **220**, power board **230**, and backlight unit **240** of FIG. **2**. It will be readily appreciated that not every method step set forth in this flow diagram is always necessary and that certain steps of the methods may be combined, performed simultaneously, in a different order, or perhaps omitted, without varying from the scope of the disclosure. While embodiments of the present disclosure are described in terms of display **200** of FIG. **2**, it should be recognized that other systems may be utilized to perform the described method.
(47) Method **800** typically starts at block **805** where scaler **210** receives an input of raw image data from display port **250**. The method proceeds to block **810** where scaler **210** may perform a color gamut enhancement algorithm as discussed in detail at FIG. **4**. The color gamut enhancement algorithm is performed for color optimization of the received raw image data generating a final image for display. The color gamut enhancement algorithm may perform the calculations every time it receives a raw image, wherein a firmware associated with scaler **210**, such as color gamut enhancement module **215**, may be updated. In another embodiment, the color gamut enhancement algorithm may query a lookup table with pre-calculated values, such as backlight intensity values, grayscale values, and pixel transmittance values. For example, color gamut enhancement algorithm module **215** may query the value for adjusting the backlight intensity from a lookup table, wherein adjusting the backlight intensity includes minimizing the backlight intensity value. Color gamut enhancement algorithm module **215** may also query the value for adjusting the grayscale setting from the lookup table. Finally, color gamut enhancement algorithm module **215** may query the value for adjusting the pixel transmittance setting from a lookup table, wherein adjusting the pixel transmittance includes maximizing the pixel transmittance value. The lookup table may have a size of 255×100 entries that may show modified grayscale and backlight intensity settings.
(48) The method proceeds to block **815** where scaler **210** may transmit the generated final image to display panel **220**. The method also proceeds to transmit a dynamic backlight PWM signal to power board **230** at block **820**. The value of the backlight PWM signal transmitted to power board **230** may change based on a desired backlight intensity and/or pixel transmittance, also referred to as liquid crystal transmittance. Determining the value of the desired pixel transmittance is discussed in detail in FIG. **4**. The method proceeds to block **830** where power board **230** may determine an output current to be transmitted to backlight unit **240**, wherein the output current is based on the value of the PWM signal received from scaler **210**.
(49) At block **840**, backlight unit **240** may provide the current received as an input to backlight elements, such as an array of light-emitting diodes. The value of the current may determine the brightness of backlight unit **240** or portions thereof which are outputted at block **850**. At block **860**, display panel **220** receives the image data along with other data for the timing controller which may be used to convert the format of the image data prior to display. At block **870**, display panel **220** may generate control signals for gate and source drivers which may be used to operate the liquid crystals in display panel **220** at block **880**. The method proceeds to block **890** where the final image is outputted at display panel **220**.
(50) Although FIG. **8** shows example blocks of method **800** in some implementations, method **800** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **8**. Those skilled in the art will understand that the principles presented herein may be implemented in any suitably arranged processing system. Additionally, or alternatively, two or more of the blocks of method **800** may be performed in parallel. For example, blocks **820** and **860** may be performed in parallel.
(51) In accordance with various embodiments of the present disclosure, the methods described

herein may be implemented by software programs executable by a computer system. Further, in an exemplary, non-limited embodiment, implementations can include distributed processing, component/object distributed processing, and parallel processing. Alternatively, virtual computer system processing can be constructed to implement one or more of the methods or functionalities as described herein.

(52) When referred to as a “device,” a “module,” a “unit,” a “controller,” or the like, the embodiments described herein can be configured as hardware. For example, a portion of an information handling system device may be hardware such as, for example, an integrated circuit (such as an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a structured ASIC, or a device embedded on a larger chip), a card (such as a Peripheral Component Interface (PCI) card, a PCI-express card, a Personal Computer Memory Card International Association (PCMCIA) card, or other such expansion card), or a system (such as a motherboard, a system-on-a-chip (SoC), or a stand-alone device).

(53) The present disclosure contemplates a computer-readable medium that includes instructions or receives and executes instructions responsive to a propagated signal; so that a device connected to a network can communicate voice, video, or data over the network. Further, the instructions may be transmitted or received over the network via the network interface device.

(54) While the computer-readable medium is shown to be a single medium, the term “computer-readable medium” includes a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. The term “computer-readable medium” shall also include any medium that is capable of storing, encoding or carrying a set of instructions for execution by a processor or that cause a computer system to perform any one or more of the methods or operations disclosed herein.

(55) In a particular non-limiting, exemplary embodiment, the computer-readable medium can include a solid-state memory such as a memory card or other package that houses one or more non-volatile read-only memories. Further, the computer-readable medium can be a random-access memory or other volatile re-writable memory. Additionally, the computer-readable medium can include a magneto-optical or optical medium, such as a disk or tapes, or another storage device to store information received via carrier wave signals such as a signal communicated over a transmission medium. A digital file attachment to an e-mail or other self-contained information archive or set of archives may be considered a distribution medium that is equivalent to a tangible storage medium. Accordingly, the disclosure is considered to include any one or more of a computer-readable medium or a distribution medium and other equivalents and successor media, in which data or instructions may be stored.

(56) Although only a few exemplary embodiments have been described in detail above, those skilled in the art will readily appreciate that many modifications are possible in the exemplary embodiments without materially departing from the novel teachings and advantages of the embodiments of the present disclosure. Accordingly, all such modifications are intended to be included within the scope of the embodiments of the present disclosure as defined in the following claims. In the claims, means-plus-function clauses are intended to cover the structures described herein as performing the recited function and not only structural equivalents but also equivalent structures.

Claims

1. A method comprising: updating, by a processor, a highest grayscale value associated with a raw image data to a maximum grayscale value; determining an expansion ratio based on the highest grayscale value and the maximum grayscale value; modifying at least one grayscale value associated with the raw image data prior to display at a display panel, wherein the modifying of the at least one grayscale value is based on the expansion ratio; and adjusting a backlight intensity

based on a quotient of the highest grayscale value divided by the maximum grayscale value, wherein the quotient is raised to a power of a gamma correction curve value to adjust a pixel transmittance of the display panel.
2. The method of claim 1, determining a pulse width modulation signal value for the adjusting of the backlight intensity.
3. The method of claim 1, wherein the adjusting of the backlight intensity includes minimizing the backlight intensity.
4. The method of claim 1, further comprising dividing the display panel into local area units.
5. The method of claim 1, further comprising dividing a backlight unit into local area units.
6. The method of claim 1, wherein the pixel transmittance is based on the gamma correction curve value.
7. The method of claim 1, wherein the pixel transmittance associated with the maximum grayscale value is one hundred percent.
8. The method of claim 1, wherein a value for the adjusting of the backlight intensity is queried from a lookup table.
9. An information handling system, comprising: a processor; and a memory storing instructions that when executed cause the processor to perform operations including: updating a highest grayscale value associated with a raw image data to a maximum grayscale value; determining an expansion ratio based on the highest grayscale value and the maximum grayscale value; modifying at least one grayscale value associated with the raw image data prior to display at a display panel, wherein the modifying of the at least one grayscale value is based on the expansion ratio; and adjusting a backlight intensity based on a quotient of the highest grayscale value divided by the maximum grayscale value, wherein the quotient is raised to a power of a gamma correction curve value to adjust a pixel transmittance of the display panel.
10. The information handling system of claim 9, wherein the operations further include determining a pulse width modulation signal value for the adjusting of the backlight intensity.
11. The information handling system of claim 9, wherein the adjusting of the backlight intensity includes minimizing the backlight intensity.
12. The information handling system of claim 9, wherein the operations further include dividing the display panel into local area units.
13. The information handling system of claim 9, wherein the adjustment of the pixel transmittance is based on the gamma correction curve value.
14. The information handling system of claim 9, wherein the pixel transmittance associated with the maximum grayscale value is one hundred percent.
15. A non-transitory computer-readable medium to store instructions that are executable to perform operations comprising: updating a highest grayscale value associated with a raw image data to a maximum grayscale value; determining an expansion ratio based on the highest grayscale value and the maximum grayscale value; modifying at least one grayscale value associated with the raw image data prior to display at a display panel, wherein the modifying of the at least one grayscale value is based on the expansion ratio; and adjusting a backlight intensity based on a quotient of the highest grayscale value divided by the maximum grayscale value, wherein the quotient is raised to a power of a gamma correction curve value to adjust a pixel transmittance of the display panel.
16. The non-transitory computer-readable medium of claim 15, wherein the operations further include determining a pulse width modulation signal value for the adjusting of the backlight intensity.
17. The non-transitory computer-readable medium of claim 15, wherein the adjusting of the backlight intensity includes minimizing the backlight intensity.
18. The non-transitory computer-readable medium of claim 15, wherein the operations further include dividing the display panel into local area units.
19. The non-transitory computer-readable medium of claim 15, wherein the operations further

include dividing a backlight unit into local area units.

20. The non-transitory computer-readable medium of claim 15, wherein the pixel transmittance is based on the gamma correction curve value.